

# Spontaneous enantiomorphism in poly-phased alkaline salts of tris(oxalato)ferrate(III): crystal structure of cubic $\text{NaRb}_5[\text{Fe}(\text{C}_2\text{O}_4)_3]_2$

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We show here that the phenomenon of spontaneous resolution of enantiomers occurs during the crystallization of the sodium and rubidium double salts of the transition metal complex tris(oxalato)ferrate(III), namely sodium penta-rubidium bis[tris(oxalato)ferrate(III)],  $\text{NaRb}_5[\text{Fe}(\text{C}_2\text{O}_4)_3]_2$ . One enantiomer of the salt crystallizes in the cubic space group  $P4_332$  with  $Z = 4$  and a Flack absolute structure parameter  $x = -0.01$  (1) and its chiral counterpart in the space group  $P4_132$  with  $x = -0.00$  (1). All metal ions are at crystallographic special positions: the iron(III) ion is on a threefold axis, coordinated by three oxalate dianions in a propeller-like conformation. One of the two independent rubidium ions is on a twofold axis in an eightfold coordination with neighbouring oxalate oxygen atoms, and the other one on a threefold axis in a sixfold  $\text{RbO}_6$  coordination. The sodium ion is at a site of  $D_3$  point group symmetry in a trigonal–antiprismatic  $\text{NaO}_6$  coordination.

## 1. Chemical context

Chirality is the structural property by which a molecule or ion cannot be superposed upon its mirror image through translation and proper rotation operations. This concept along with the related ones of chiral crystal structures and space groups is discussed by Flack (2003). Chirality is at the core (among other research areas) of the not yet understood origin of the biomolecular asymmetry of life (Meierhenrich, 2008), enantioselective chemical reactions (Knowles, 2001; Noyori, 2001; Sharpless, 2001), biological activity of pharmaceuticals (Nguyen *et al.*, 2006) and in the design of multifunctional solid-state materials endowed with optical activity and long-range magnetic order (Coronado *et al.*, 2003) and also in the understanding of the physical properties of chiral liquid crystals and their tailoring for applications in opto-electronic devices (Goodby, 1998; Coles, 1998).

While attempting to crystallize the rubidium salt of the tris(oxalato)ferrate(III) transition metal complex, one of the preparations segregated into a poly-phased crystal system. It contained the intended  $\text{Rb}_3[\text{Fe}(\text{C}_2\text{O}_4)_3] \cdot 3\text{H}_2\text{O}$  compound (monoclinic  $P2_1/c$ ), which turned out to be isotypic to the reported potassium salt (Junk, 2005; Piro *et al.*, 2016), and the triclinic ( $P\bar{1}$ )  $\text{Rb}(\text{C}_2\text{O}_4\text{H})(\text{C}_2\text{O}_4\text{H}_2) \cdot 2\text{H}_2\text{O}$  salt (Kherfi *et al.*, 2010), which is isotypic to the ammonium analogue (Jarzemska *et al.*, 2014). A third phase consisted of large green crystals of a new cubic ( $P4_332$ )  $\text{NaRb}_5[\text{Fe}(\text{C}_2\text{O}_4)_3]_2$  salt. Interestingly, the isotypic counterpart of this salt where rubi-

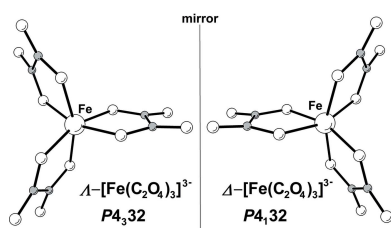


Table 1

Bond lengths and angles ( $\text{\AA}$ ,  $^\circ$ ) around iron(III) and within the oxalate dianion in  $\text{NaRb}_5[\text{Fe}(\text{C}_2\text{O}_4)_3]_2$   $P4_332$  enantiomer.

(a) At a crystal site of  $C_3$  point group symmetry.

| Iron(III) <sup>a</sup>   |           | $(\text{C}_2\text{O}_4)^{2-}$ |           |
|--------------------------|-----------|-------------------------------|-----------|
| Fe—O11                   | 2.021 (4) | C1—O12                        | 1.211 (7) |
| Fe—O21                   | 1.989 (4) | C1—O11                        | 1.286 (7) |
|                          |           | C1—C2                         | 1.540 (9) |
| O21—Fe—O11               | 80.0 (2)  | C2—O22                        | 1.211 (7) |
| O21—Fe—O21 <sup>i</sup>  | 88.4 (2)  | C2—O21                        | 1.283 (7) |
| O11—Fe—O11 <sup>i</sup>  | 88.7 (2)  |                               |           |
| O11—Fe—O21 <sup>i</sup>  | 106.2 (2) | O12—C—O11                     | 125.2 (6) |
| O11—Fe—O21 <sup>ii</sup> | 160.9 (2) | O12—C1—C2                     | 121.2 (6) |
|                          |           | O11—C1—C2                     | 113.5 (5) |
|                          |           | O22—C2—O21                    | 125.3 (6) |
|                          |           | O22—C2—C1                     | 121.1 (6) |
|                          |           | O21—C2—C1                     | 113.6 (5) |

Symmetry codes: (i)  $-z + \frac{1}{2}, -x + 1, y - \frac{1}{2}$ ; (ii)  $-y + 1, z + \frac{1}{2}, -x + \frac{1}{2}$ .

dium is replaced by potassium has been reported by Wartchow (1997) to appear in a mixture with crystals of the monoclinic  $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3] \cdot 3\text{H}_2\text{O}$  salt, hence confirming the tendency of potassium and rubidium alkaline ions to form isotopic crystal analogues (Piro *et al.*, 2016). Curiously, in a previous work, Henneicke & Wartchow (1997) reported the chiral counterpart of the cubic  $\text{NaK}_5[\text{Fe}(\text{C}_2\text{O}_4)_3]_2$  salt, which crystallizes in the space group  $P4_332$ . This prompted us to search for the chiral rubidium analogue in the very same batch as the single-crystals that solved in the space group  $P4_332$   $\text{NaRb}_5[\text{Fe}(\text{C}_2\text{O}_4)_3]_2$ . By chance, we picked a single crystal and submitted it to X-ray diffraction scrutiny to find that it now belonged to the chiral space group  $P4_332$ . This strongly suggests that the  $\text{NaM}_5[\text{Fe}(\text{C}_2\text{O}_4)_3]_2$  ( $M = \text{K}, \text{Rb}$ ) crystal samples could be racemic conglomerates generated by spontaneous resolution, a rare event discovered by Louis Pasteur in 1848 (Pasteur, 1848*a,b*) in a famous experiment in which he hand-sorted the chirally resolved crystals of sodium ammonium tartrate tetrahydrate on the basis of their observed morphology and then examined their respective solutions with a polarimeter to find opposite rotations of the plane of light polarization (Flack, 2009). Recently, we found that the phenomenon could also have occurred in isotopic  $[\text{M}(\text{Lap})_2]_n$  ( $M = \text{Cd}, \text{Mn}$ ; HLap = 2-hydroxy-3-(3-methyl-2-butenyl)-1,4-naphthoquinone,  $\text{C}_{15}\text{H}_{14}\text{O}_3$ ) complexes whose enantiomers crystallize in the tetragonal and enantiomorphic space groups  $P4_32_12$  and  $P4_12_12$  (Farfán *et al.*, 2015).

## 2. Structural commentary

Fig. 1 shows an ORTEP (Farrugia, 2012) drawing of the  $P4_332$  enantiomer of the title compound. Bond lengths and angles around iron(III) and within the oxalate dianion are listed in Table 1 and contact distances around the alkali ions are shown in Table 2. All metal ions are at crystallographic special positions while the oxalate anion is on a general position. The iron(III) ion is on a threefold axis,  $C_3$  point group symmetry (Wyckoff  $c$  site), in an octahedral environment ( $\text{FeO}_6$  core). It is coordinated to three, symmetry-related, oxalate anions acting as bidentate ligands through the oxygen atoms of their opposite carboxylic groups in a propeller-like conformation and along one electron pair lobe on each oxygen ligand. The

$\text{FeO}_6$  bond geometry and metrics are consistent with the oxalate being a weak-field ligand that gives rise to the high-spin ( $S = 5/2$ ) electronic ground state exhibited by the complex, as probed by magnetic susceptibility (Delgado *et al.*, 2002) and ESR spectroscopy (Collison & Powell, 1990) in synthetic minguzzite,  $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3] \cdot 3\text{H}_2\text{O}$ , by polarized electronic absorption spectroscopy in single crystal  $\text{NaMg}[(\text{Fe}, \text{Al})(\text{C}_2\text{O}_4)_3] \cdot 9\text{H}_2\text{O}$  mixtures (Piper & Carlin, 1961) and also by

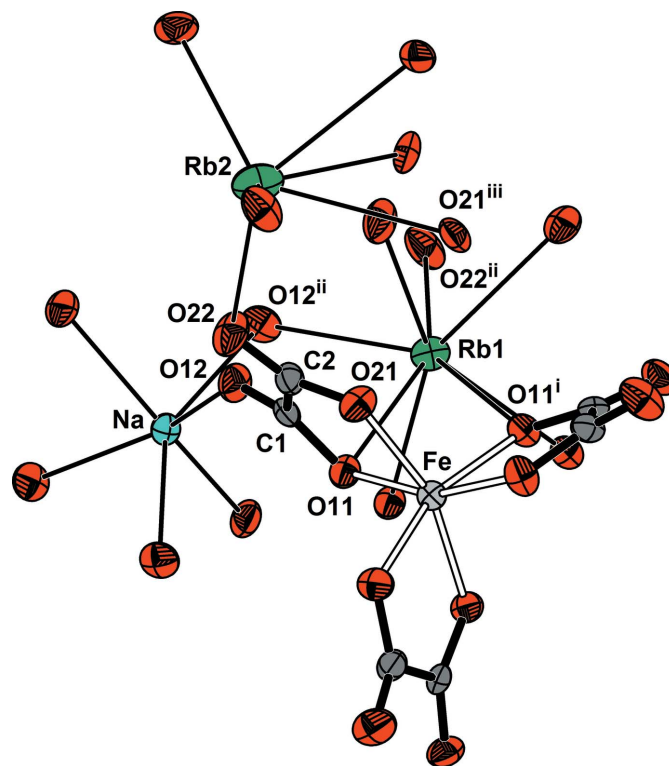


Figure 1

View of  $\text{NaRb}_5[\text{Fe}(\text{C}_2\text{O}_4)_3]_2$  showing the atom labels and displacement ellipsoids at the 50% probability level. For clarity, only the minimum number of oxygen ligands around each metal ion has been labelled. The rest of the environmental oxygen atoms are generated through the symmetry operations of the corresponding point groups:  $C_3$  (Fe),  $C_2$  (Rb1),  $C_3$  (Rb2) and  $D_3$  (Na). Iron–oxalate bonds are indicated by double lines and alkali metal–oxygen contacts by single lines. Symmetry codes: (i)  $-y + 1, z + \frac{1}{2}, -x + \frac{1}{2}$ ; (ii)  $y - \frac{1}{2}, -z + \frac{3}{2}, -x + 1$ ; (iii)  $-y + \frac{5}{4}, -x + \frac{5}{4}, -z + \frac{1}{4}$ .

Table 2

Bond lengths (Å) around the alkali metal ions in  $\text{NaRb}_5[\text{Fe}(\text{C}_2\text{O}_4)_3]_2$   $P4_332$  enantiomer..

(a) At a site of  $C_2$  point group symmetry; (b) at a  $C_3$  site; (c) at a  $D_3$  site.

| Rb1 <sup>a</sup>      |           | Rb2 <sup>b</sup>       |           | Na <sup>c</sup> |           |
|-----------------------|-----------|------------------------|-----------|-----------------|-----------|
| Rb1—O11               | 3.009 (4) | Rb2—O22                | 2.808 (4) | Na—O12          | 2.439 (4) |
| Rb1—O11 <sup>i</sup>  | 3.067 (4) | Rb2—O21 <sup>iii</sup> | 3.114 (4) |                 |           |
| Rb1—O22 <sup>ii</sup> | 2.788 (5) |                        |           |                 |           |
| Rb1—O12 <sup>ii</sup> | 3.133 (5) |                        |           |                 |           |

Symmetry codes: (i)  $-y + 1, z + \frac{1}{2}, -x + \frac{1}{2}$ ; (ii)  $y - \frac{1}{2}, -z + \frac{3}{2}, -x + 1$ ; (iii)  $-y + \frac{5}{4}, -x + \frac{5}{4}, -z + \frac{1}{4}$ .

Mössbauer spectroscopy in  $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3] \cdot 3\text{H}_2\text{O}$  (Bancroft *et al.*, 1970; Sato & Tominaga, 1979; Ladriere, 1992) and in the alkali (Na, Rb, Cs) family of tris(oxalato)ferrate(III) salts (Piro *et al.*, 2016).

The planes of the carboxylic  $-\text{COO}^-$  groups of the oxalate ligand are slightly tilted from each other, by  $12 (1)^\circ$  around the C—C  $\sigma$ -bond. As expected, the C—O bond lengths involving the coordinated-to-metal oxygen atoms are significantly longer [1.286 (7) and 1.283 (7) Å] than the ones corresponding to the uncoordinated oxalate oxygen atoms [both equal to 1.211 (7) Å].

There are two independent rubidium ions, one (Rb1) lying on a twofold axis,  $C_2$  point group symmetry (*d* site) in an eightfold coordination with neighbouring oxalate oxygen atoms, the other one (Rb2) on a threefold axis,  $C_3$  point group (*c* site) in a sixfold coordination. The sodium ion is at a site of  $D_3$  point group symmetry (*a* site) in a trigonal-antiprismatic  $\text{NaO}_6$  coordination with one oxygen atom of six neighbouring, symmetry-related, oxalate ions.

When dealing with octahedral  $\text{Fe}(\text{C}_2\text{O}_4)_3$  tris-chelated metal complexes, it is customary to describe its chirality employing  $\Lambda$ - and  $\Delta$ -descriptors (Meierhenrich, 2008). It turns out that the enantiomeric complexes correlate with the corresponding chiral space groups, as indicated in Fig. 2.

The possibility of controlling the crystal chirality and therefore obtaining enhanced optical activity of functional materials has been discussed (Gruselle *et al.*, 2006). To this purpose, two general synthetic routes have been developed to

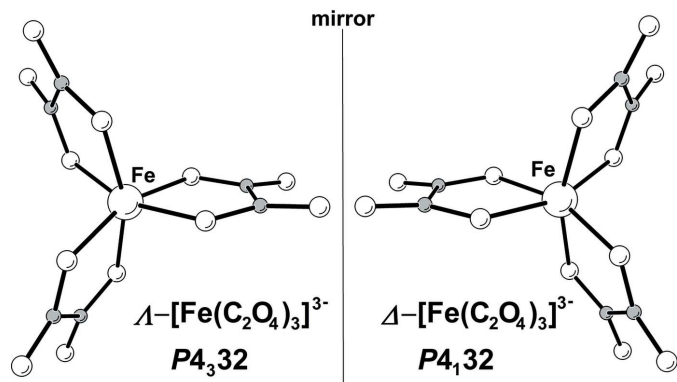


Figure 2  
Views of the  $\Lambda$  and  $\Delta$  enantiomers of  $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ .

reach optically active coordination compounds, namely either by enantioselective synthesis using enantiopure chiral species, which yields enantiopure samples (Knof & von Zelewsky, 1999) or by spontaneous resolution upon crystallization from a racemate, which yields a conglomerate (Pérez-García & Amabilino, 2002). As explained above, the chiral  $\text{NaRb}_5[\text{Fe}(\text{C}_2\text{O}_4)_3]_2$  crystals were obtained through the phenomena of spontaneous resolution from a racemic solution of  $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$  complex ions into a racemic conglomerate. This is presumably followed by a structural inductive effect by these chiral molecular ions on the alkali metal ions through shared oxalate ligands. In fact, not only is the  $\text{Fe}^{\text{III}}$  ion a 'stereogenic centre' in the  $\text{Fe}(\text{C}_2\text{O}_4)_3$  tris-chelated metal complex, but so also are the sodium and one (Rb2) of the rubidium ions. These metal ions are in a distorted octahedral environment coordinated to six oxalate anions, acting as monodentate ligand through one of their oxygen atoms and resembling a six-bladed propeller-like conformation. From the structural data, it turns out that the chirality of this local arrangement around the alkaline ions is coincident with the one of the  $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$  inductor and therefore the chiral crystals reported here can be more conveniently described as  $\Lambda\text{-Na}\Lambda\text{-Rb}_2\text{Rb}_3[\Lambda\text{-Fe}(\text{C}_2\text{O}_4)_3]_2$  ( $P4_332$ ) and  $\Delta\text{-Na}\Delta\text{-Rb}_2\text{Rb}_3[\Delta\text{-Fe}(\text{C}_2\text{O}_4)_3]_2$  ( $P4_132$ ). However, no definitive chirality can be unambiguously assigned to the other independent rubidium (Rb1) ion which is in an eightfold polyhedral coordination.

### 3. Database survey

The formation of racemic conglomerates of single crystals, adequate for structural X-ray diffraction, generated by spontaneous resolution is an infrequent phenomenon. In fact, a search of the Cambridge Structural Database (Groom *et al.*, 2016) invoking the term 'spontaneous resolution' showed seventeen entries, and another one using as a target 'chiral crystals' produced a further four hits. Among them there were reported the chiral to each other (*M*)- and (*P*)-catena- $\{[\mu_2\text{-}2\text{-(imidazo[4,5-f](1,10)phenanthroline-2-yl)benzoato-}N,N',O\text{-aquachlorozinc(II)}]\}$  (CSD refcodes EJINOB and EJINUH; Wei *et al.*, 2011) and catena- $\{[\mu_8\text{-benzene-1,3,5-tricarboxylato}]\text{lithiumzinc}\}$  (CSD refcodes WJHUM and WJJAU; Xie *et al.*, 2010).

### 4. Synthesis and crystallization

As stated in the *Chemical context*, in one of the preparations generated during the synthesis of the rubidium salt of  $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ , by reaction of freshly precipitated  $\text{Fe}(\text{OH})_3$  (obtained by dropwise addition of a small excess of 20% NaOH to an  $\text{Fe}^{\text{III}}$  solution) with rubidium oxalate:  $\text{Fe}(\text{OH})_3 + 3\text{Rb}(\text{HC}_2\text{O}_4) + 3\text{H}_2\text{O} \rightarrow \text{Rb}_3[\text{Fe}(\text{C}_2\text{O}_4)_3] \cdot 3\text{H}_2\text{O} + 3\text{H}_2\text{O}$  (Piro *et al.*, 2016), we found a relatively complex reaction giving rise to a poly-phased crystal mixture, from which the  $\text{NaRb}_5[\text{Fe}(\text{C}_2\text{O}_4)_3]_2$  chiral pair could be isolated.

**Table 3**  
Experimental details.

|                                                                            | Cubic, $P4_332$                                                                                                    | Cubic, $P4_132$                                                                                                     |
|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Crystal data                                                               |                                                                                                                    |                                                                                                                     |
| Chemical formula                                                           | $\text{NaRb}_5[\text{Fe}(\text{C}_2\text{O}_4)_3]_2$                                                               | $\text{NaRb}_5[\text{Fe}(\text{C}_2\text{O}_4)_3]_2$                                                                |
| $M_r$                                                                      | 1090.16                                                                                                            | 1090.16                                                                                                             |
| Temperature (K)                                                            | 297                                                                                                                | 293                                                                                                                 |
| $a$ (Å)                                                                    | 13.8058 (4)                                                                                                        | 13.7995 (3)                                                                                                         |
| $V$ (Å <sup>3</sup> )                                                      | 2631.4 (2)                                                                                                         | 2627.79 (17)                                                                                                        |
| $Z$                                                                        | 4                                                                                                                  | 4                                                                                                                   |
| Radiation type                                                             | Mo $K\alpha$                                                                                                       | Mo $K\alpha$                                                                                                        |
| $\mu$ (mm <sup>-1</sup> )                                                  | 10.42                                                                                                              | 10.43                                                                                                               |
| Crystal size (mm)                                                          | 0.48 × 0.42 × 0.38                                                                                                 | 0.48 × 0.35 × 0.25                                                                                                  |
| Data collection                                                            |                                                                                                                    |                                                                                                                     |
| Diffractometer                                                             | Agilent Xcalibur Eos Gemini                                                                                        | Rigaku Oxford Diffraction Xcalibur, Eos, Gemini                                                                     |
| Absorption correction                                                      | Multi-scan ( <i>CrysAlis PRO</i> ; Agilent, 2014)                                                                  | Multi-scan ( <i>CrysAlis PRO</i> ; Rigaku OD, 2015)                                                                 |
| $T_{\min}$ , $T_{\max}$                                                    | 0.690, 1.000                                                                                                       | 0.786, 1.000                                                                                                        |
| No. of measured, independent and observed [ $I > 2\sigma(I)$ ] reflections | 2960, 959, 767                                                                                                     | 4284, 961, 814                                                                                                      |
| $R_{\text{int}}$                                                           | 0.043                                                                                                              | 0.038                                                                                                               |
| $(\sin \theta/\lambda)_{\text{max}}$ (Å <sup>-1</sup> )                    | 0.638                                                                                                              | 0.638                                                                                                               |
| Refinement                                                                 |                                                                                                                    |                                                                                                                     |
| $R[F^2 > 2\sigma(F^2)]$ , $wR(F^2)$ , $S$                                  | 0.035, 0.064, 1.00                                                                                                 | 0.032, 0.068, 1.02                                                                                                  |
| No. of reflections                                                         | 959                                                                                                                | 961                                                                                                                 |
| No. of parameters                                                          | 68                                                                                                                 | 68                                                                                                                  |
| $\Delta\rho_{\text{max}}$ , $\Delta\rho_{\text{min}}$ (e Å <sup>-3</sup> ) | 0.84, -0.85                                                                                                        | 1.02, -0.95                                                                                                         |
| Absolute structure                                                         | Flack $x$ determined using 225 quotients<br>[ $(I^+) - (I^-)$ ]/[ $(I^+) + (I^-)$ ] (Parsons <i>et al.</i> , 2013) | Flack $x$ determined using 251 quotients<br>[ $(I^+) - (I^-)$ ]/[ $(I^+) + (I^-)$ ] (Parsons <i>et al.</i> , 2013). |
| Absolute structure parameter                                               | -0.013 (12)                                                                                                        | -0.003 (10)                                                                                                         |

Computer programs: *CrysAlis PRO* (Agilent, 2014; Rigaku OD, 2015), *SHELXT* (Sheldrick, 2015a), *SHELXL2014* (Sheldrick, 2015b) and *ORTEP-3 for Windows* (Farrugia, 2012).

## 5. Refinement details

Crystal data, data collection procedure and structure refinement results are summarized in Table 3. The structure was solved by intrinsic phasing with *SHELXT* (Sheldrick, 2015a). The stereoisomers were determined through refinement of the Flack absolute structure parameter. This is the fractional contribution to the diffraction pattern due to the molecule racemic twin and for the correct enantiomeric crystal it should be zero to within experimental error.

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## supporting information

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## Spontaneous enantiomorphism in poly-phased alkaline salts of tris-(oxalato)ferrate(III): crystal structure of cubic $\text{NaRb}_5[\text{Fe}(\text{C}_2\text{O}_4)_3]_2$

O. E. Piro, G. A. Echeverría and E. J. Baran

### Computing details

Data collection: *CrysAlis PRO* (Agilent, 2014) for P4332; *CrysAlis PRO* (Rigaku OD, 2015) for P4132. Cell refinement: *CrysAlis PRO* (Agilent, 2014) for P4332; *CrysAlis PRO* (Rigaku OD, 2015) for P4132. Data reduction: *CrysAlis PRO* (Agilent, 2014) for P4332; *CrysAlis PRO* (Rigaku OD, 2015) for P4132. For both structures, program(s) used to solve structure: SHELXT (Sheldrick, 2015a); program(s) used to refine structure: *SHELXL2014* (Sheldrick, 2015b); molecular graphics: *ORTEP-3 for Windows* (Farrugia, 2012); software used to prepare material for publication: *SHELXL2014* (Sheldrick, 2015b).

### Sodium pentarubidium bis[tris(oxalato)ferrate(III)] (P4332)

#### Crystal data

$\text{NaRb}_5[\text{Fe}(\text{C}_2\text{O}_4)_3]_2$

$M_r = 1090.16$

Cubic,  $P4_332$

$a = 13.8058(4) \text{ \AA}$

$V = 2631.4(2) \text{ \AA}^3$

$Z = 4$

$F(000) = 2048$

$D_x = 2.752 \text{ Mg m}^{-3}$

Mo  $K\alpha$  radiation,  $\lambda = 0.71073 \text{ \AA}$

Cell parameters from 922 reflections

$\theta = 3.6\text{--}27.3^\circ$

$\mu = 10.42 \text{ mm}^{-1}$

$T = 297 \text{ K}$

Fragment, green

$0.48 \times 0.42 \times 0.38 \text{ mm}$

#### Data collection

Agilent Xcalibur Eos Gemini  
diffractometer

Radiation source: Enhance (Mo) X-ray Source

Graphite monochromator

Detector resolution:  $16.0604 \text{ pixels mm}^{-1}$

$\omega$  scans

Absorption correction: multi-scan  
(*CrysAlis PRO*; Agilent, 2014)

$T_{\min} = 0.690$ ,  $T_{\max} = 1.000$

2960 measured reflections

959 independent reflections

767 reflections with  $I > 2\sigma(I)$

$R_{\text{int}} = 0.043$

$\theta_{\max} = 27.0^\circ$ ,  $\theta_{\min} = 3.3^\circ$

$h = -8 \rightarrow 17$

$k = -11 \rightarrow 16$

$l = -8 \rightarrow 10$

#### Refinement

Refinement on  $F^2$

Least-squares matrix: full

$R[F^2 > 2\sigma(F^2)] = 0.035$

$wR(F^2) = 0.064$

$S = 1.00$

959 reflections

68 parameters

0 restraints

Primary atom site location: dual

$w = 1/[\sigma^2(F_o^2) + (0.0274P)^2]$

where  $P = (F_o^2 + 2F_c^2)/3$

$(\Delta/\sigma)_{\max} < 0.001$

$\Delta\rho_{\max} = 0.84 \text{ e \AA}^{-3}$

$\Delta\rho_{\min} = -0.85 \text{ e \AA}^{-3}$



Absolute structure: Flack  $x$  determined using  
 225 quotients  $[(F^+)-(F)]/[(F^+)+(F)]$  (Parsons *et al.*,  
 2013)  
 Absolute structure parameter:  $-0.013$  (12)

### Special details

**Geometry.** All esds (except the esd in the dihedral angle between two l.s. planes) are estimated using the full covariance matrix. The cell esds are taken into account individually in the estimation of esds in distances, angles and torsion angles; correlations between esds in cell parameters are only used when they are defined by crystal symmetry. An approximate (isotropic) treatment of cell esds is used for estimating esds involving l.s. planes.

### Fractional atomic coordinates and isotropic or equivalent isotropic displacement parameters ( $\text{\AA}^2$ )

|     | $x$         | $y$         | $z$         | $U_{\text{iso}}^*/U_{\text{eq}}$ |
|-----|-------------|-------------|-------------|----------------------------------|
| C1  | 0.3861 (5)  | 0.8068 (4)  | 0.3863 (5)  | 0.0207 (13)                      |
| C2  | 0.4510 (5)  | 0.7715 (5)  | 0.3027 (5)  | 0.0234 (16)                      |
| O11 | 0.2953 (3)  | 0.7918 (3)  | 0.3718 (3)  | 0.0241 (10)                      |
| O12 | 0.4203 (3)  | 0.8466 (3)  | 0.4566 (3)  | 0.0312 (11)                      |
| O21 | 0.4037 (3)  | 0.7481 (3)  | 0.2262 (3)  | 0.0287 (10)                      |
| O22 | 0.5380 (3)  | 0.7665 (4)  | 0.3119 (3)  | 0.0402 (13)                      |
| Fe  | 0.26048 (6) | 0.73952 (6) | 0.23952 (6) | 0.0209 (4)                       |
| Rb1 | 0.25612 (5) | 1.00612 (5) | 0.3750      | 0.0290 (3)                       |
| Rb2 | 0.67150 (6) | 0.82850 (6) | 0.17150 (6) | 0.0444 (4)                       |
| Na  | 0.3750      | 0.8750      | 0.6250      | 0.0218 (13)                      |

### Atomic displacement parameters ( $\text{\AA}^2$ )

|     | $U^{11}$    | $U^{22}$    | $U^{33}$    | $U^{12}$      | $U^{13}$       | $U^{23}$       |
|-----|-------------|-------------|-------------|---------------|----------------|----------------|
| C1  | 0.025 (3)   | 0.020 (3)   | 0.017 (3)   | $-0.001$ (3)  | $-0.001$ (3)   | 0.006 (3)      |
| C2  | 0.021 (3)   | 0.029 (4)   | 0.020 (3)   | 0.002 (3)     | 0.000 (3)      | 0.004 (3)      |
| O11 | 0.020 (2)   | 0.032 (3)   | 0.020 (2)   | 0.000 (2)     | 0.002 (2)      | 0.000 (2)      |
| O12 | 0.034 (3)   | 0.041 (3)   | 0.019 (2)   | $-0.003$ (2)  | $-0.004$ (2)   | $-0.002$ (2)   |
| O21 | 0.023 (2)   | 0.044 (3)   | 0.019 (2)   | $-0.004$ (2)  | 0.006 (2)      | $-0.006$ (2)   |
| O22 | 0.020 (2)   | 0.069 (4)   | 0.032 (3)   | 0.004 (3)     | 0.003 (2)      | 0.002 (3)      |
| Fe  | 0.0209 (4)  | 0.0209 (4)  | 0.0209 (4)  | $-0.0006$ (4) | $-0.0006$ (4)  | 0.0006 (4)     |
| Rb1 | 0.0290 (3)  | 0.0290 (3)  | 0.0288 (5)  | $-0.0018$ (5) | 0.0051 (3)     | $-0.0051$ (3)  |
| Rb2 | 0.0444 (4)  | 0.0444 (4)  | 0.0444 (4)  | $-0.0115$ (4) | 0.0115 (4)     | $-0.0115$ (4)  |
| Na  | 0.0218 (13) | 0.0218 (13) | 0.0218 (13) | 0.0014 (16)   | $-0.0014$ (16) | $-0.0014$ (16) |

### Geometric parameters ( $\text{\AA}$ , $^\circ$ )

|                      |           |                         |           |
|----------------------|-----------|-------------------------|-----------|
| C1—O12               | 1.211 (7) | Rb1—O22 <sup>v</sup>    | 2.788 (5) |
| C1—O11               | 1.286 (7) | Rb1—O22 <sup>vi</sup>   | 2.788 (5) |
| C1—C2                | 1.540 (9) | Rb1—O11 <sup>vii</sup>  | 3.009 (4) |
| C2—O22               | 1.211 (7) | Rb1—O11 <sup>iv</sup>   | 3.067 (4) |
| C2—O21               | 1.283 (7) | Rb1—O11 <sup>viii</sup> | 3.067 (4) |
| O11—Fe               | 2.021 (4) | Rb1—O12 <sup>v</sup>    | 3.133 (5) |
| O11—Rb1              | 3.009 (4) | Rb1—O12 <sup>vi</sup>   | 3.133 (5) |
| O11—Rb1 <sup>i</sup> | 3.067 (4) | Rb1—O12 <sup>vii</sup>  | 3.354 (5) |

|                                         |             |                                             |             |
|-----------------------------------------|-------------|---------------------------------------------|-------------|
| O12—Na                                  | 2.439 (4)   | Rb2—O22 <sup>ix</sup>                       | 2.808 (4)   |
| O12—Rb1 <sup>ii</sup>                   | 3.133 (5)   | Rb2—O22 <sup>x</sup>                        | 2.808 (4)   |
| O12—Rb1                                 | 3.354 (5)   | Rb2—O21 <sup>iii</sup>                      | 3.114 (4)   |
| O21—Fe                                  | 1.989 (4)   | Rb2—O21 <sup>xi</sup>                       | 3.114 (4)   |
| O21—Rb2 <sup>iii</sup>                  | 3.114 (4)   | Rb2—O21 <sup>vi</sup>                       | 3.114 (4)   |
| O22—Rb1 <sup>ii</sup>                   | 2.788 (5)   | Na—O12 <sup>xii</sup>                       | 2.439 (4)   |
| O22—Rb2                                 | 2.808 (4)   | Na—O12 <sup>ii</sup>                        | 2.439 (4)   |
| Fe—O21 <sup>iv</sup>                    | 1.989 (4)   | Na—O12 <sup>xiii</sup>                      | 2.439 (4)   |
| Fe—O21 <sup>i</sup>                     | 1.989 (4)   | Na—O12 <sup>viii</sup>                      | 2.439 (4)   |
| Fe—O11 <sup>i</sup>                     | 2.021 (4)   | Na—O12 <sup>v</sup>                         | 2.439 (4)   |
| Fe—O11 <sup>iv</sup>                    | 2.021 (4)   |                                             |             |
| O12—C1—O11                              | 125.2 (6)   | O11 <sup>vii</sup> —Rb1—O12 <sup>v</sup>    | 110.45 (11) |
| O12—C1—C2                               | 121.2 (6)   | O11 <sup>iv</sup> —Rb1—O12 <sup>v</sup>     | 151.29 (11) |
| O11—C1—C2                               | 113.5 (5)   | O11 <sup>viii</sup> —Rb1—O12 <sup>v</sup>   | 65.47 (11)  |
| O22—C2—O21                              | 125.3 (6)   | O22 <sup>v</sup> —Rb1—O12 <sup>vi</sup>     | 91.18 (12)  |
| O22—C2—C1                               | 121.1 (6)   | O22 <sup>vi</sup> —Rb1—O12 <sup>vi</sup>    | 56.14 (12)  |
| O21—C2—C1                               | 113.6 (5)   | O11—Rb1—O12 <sup>vi</sup>                   | 110.45 (11) |
| C1—O11—Fe                               | 115.5 (4)   | O11 <sup>vii</sup> —Rb1—O12 <sup>vi</sup>   | 95.92 (11)  |
| C1—O11—Rb1                              | 90.8 (3)    | O11 <sup>iv</sup> —Rb1—O12 <sup>vi</sup>    | 65.47 (11)  |
| Fe—O11—Rb1                              | 108.75 (16) | O11 <sup>viii</sup> —Rb1—O12 <sup>vi</sup>  | 151.29 (11) |
| C1—O11—Rb1 <sup>i</sup>                 | 133.5 (4)   | O12 <sup>v</sup> —Rb1—O12 <sup>vi</sup>     | 133.07 (16) |
| Fe—O11—Rb1 <sup>i</sup>                 | 106.67 (16) | O22 <sup>v</sup> —Rb1—O12 <sup>vii</sup>    | 64.52 (12)  |
| Rb1—O11—Rb1 <sup>i</sup>                | 93.73 (11)  | O22 <sup>vi</sup> —Rb1—O12 <sup>vii</sup>   | 114.33 (13) |
| C1—O12—Na                               | 137.5 (4)   | O11—Rb1—O12 <sup>vii</sup>                  | 141.04 (11) |
| C1—O12—Rb1 <sup>ii</sup>                | 107.6 (4)   | O11 <sup>vii</sup> —Rb1—O12 <sup>vii</sup>  | 40.34 (11)  |
| Na—O12—Rb1 <sup>ii</sup>                | 97.96 (14)  | O11 <sup>iv</sup> —Rb1—O12 <sup>vii</sup>   | 90.55 (10)  |
| C1—O12—Rb1                              | 76.5 (4)    | O11 <sup>viii</sup> —Rb1—O12 <sup>vii</sup> | 90.35 (11)  |
| Na—O12—Rb1                              | 92.39 (13)  | O12 <sup>v</sup> —Rb1—O12 <sup>vii</sup>    | 116.99 (13) |
| Rb1 <sup>ii</sup> —O12—Rb1              | 157.89 (15) | O12 <sup>vi</sup> —Rb1—O12 <sup>vii</sup>   | 62.32 (15)  |
| C2—O21—Fe                               | 116.4 (4)   | O22 <sup>v</sup> —Rb1—O12                   | 114.33 (14) |
| C2—O21—Rb2 <sup>iii</sup>               | 133.5 (4)   | O22 <sup>vi</sup> —Rb1—O12                  | 64.52 (12)  |
| Fe—O21—Rb2 <sup>iii</sup>               | 95.43 (15)  | O11—Rb1—O12                                 | 40.34 (11)  |
| C2—O22—Rb1 <sup>ii</sup>                | 121.7 (4)   | O11 <sup>vii</sup> —Rb1—O12                 | 141.04 (11) |
| C2—O22—Rb2                              | 124.2 (4)   | O11 <sup>iv</sup> —Rb1—O12                  | 90.35 (11)  |
| Rb1 <sup>ii</sup> —O22—Rb2              | 110.80 (17) | O11 <sup>viii</sup> —Rb1—O12                | 90.55 (10)  |
| O21 <sup>iv</sup> —Fe—O21 <sup>i</sup>  | 88.42 (18)  | O12 <sup>v</sup> —Rb1—O12                   | 62.32 (15)  |
| O21 <sup>iv</sup> —Fe—O21               | 88.43 (18)  | O12 <sup>vi</sup> —Rb1—O12                  | 116.99 (13) |
| O21 <sup>i</sup> —Fe—O21                | 88.42 (18)  | O12 <sup>vii</sup> —Rb1—O12                 | 178.45 (16) |
| O21 <sup>iv</sup> —Fe—O11               | 160.91 (17) | O22 <sup>ix</sup> —Rb2—O22 <sup>x</sup>     | 116.27 (7)  |
| O21 <sup>i</sup> —Fe—O11                | 106.20 (17) | O22 <sup>ix</sup> —Rb2—O22                  | 116.27 (7)  |
| O21—Fe—O11                              | 79.96 (16)  | O22 <sup>x</sup> —Rb2—O22                   | 116.27 (7)  |
| O21 <sup>iv</sup> —Fe—O11 <sup>i</sup>  | 106.20 (17) | O22 <sup>ix</sup> —Rb2—O21 <sup>iii</sup>   | 131.79 (14) |
| O21 <sup>i</sup> —Fe—O11 <sup>i</sup>   | 79.96 (16)  | O22 <sup>x</sup> —Rb2—O21 <sup>iii</sup>    | 81.60 (13)  |
| O21—Fe—O11 <sup>i</sup>                 | 160.91 (17) | O22—Rb2—O21 <sup>iii</sup>                  | 89.01 (12)  |
| O11—Fe—O11 <sup>i</sup>                 | 88.73 (17)  | O22 <sup>ix</sup> —Rb2—O21 <sup>xi</sup>    | 81.60 (13)  |
| O21 <sup>iv</sup> —Fe—O11 <sup>iv</sup> | 79.96 (16)  | O22 <sup>x</sup> —Rb2—O21 <sup>xi</sup>     | 89.01 (12)  |
| O21 <sup>i</sup> —Fe—O11 <sup>iv</sup>  | 160.91 (17) | O22—Rb2—O21 <sup>xi</sup>                   | 131.79 (14) |



|                                             |             |                                              |              |
|---------------------------------------------|-------------|----------------------------------------------|--------------|
| O21—Fe—O11 <sup>iv</sup>                    | 106.20 (17) | O21 <sup>iii</sup> —Rb2—O21 <sup>xi</sup>    | 52.88 (13)   |
| O11—Fe—O11 <sup>iv</sup>                    | 88.72 (17)  | O22 <sup>ix</sup> —Rb2—O21 <sup>vi</sup>     | 89.01 (12)   |
| O11 <sup>i</sup> —Fe—O11 <sup>iv</sup>      | 88.73 (17)  | O22 <sup>x</sup> —Rb2—O21 <sup>vi</sup>      | 131.79 (14)  |
| O22 <sup>v</sup> —Rb1—O22 <sup>vi</sup>     | 95.3 (2)    | O22—Rb2—O21 <sup>vi</sup>                    | 81.60 (13)   |
| O22 <sup>v</sup> —Rb1—O11                   | 152.05 (12) | O21 <sup>iii</sup> —Rb2—O21 <sup>vi</sup>    | 52.88 (13)   |
| O22 <sup>vi</sup> —Rb1—O11                  | 83.24 (13)  | O21 <sup>xi</sup> —Rb2—O21 <sup>vi</sup>     | 52.88 (13)   |
| O22 <sup>v</sup> —Rb1—O11 <sup>vii</sup>    | 83.24 (13)  | O12—Na—O12 <sup>xii</sup>                    | 76.1 (2)     |
| O22 <sup>vi</sup> —Rb1—O11 <sup>vii</sup>   | 152.05 (12) | O12—Na—O12 <sup>ii</sup>                     | 87.13 (15)   |
| O11—Rb1—O11 <sup>vii</sup>                  | 110.74 (16) | O12 <sup>xii</sup> —Na—O12 <sup>ii</sup>     | 120.9 (2)    |
| O22 <sup>v</sup> —Rb1—O11 <sup>iv</sup>     | 152.40 (12) | O12—Na—O12 <sup>xiii</sup>                   | 145.7 (2)    |
| O22 <sup>vi</sup> —Rb1—O11 <sup>iv</sup>    | 83.98 (12)  | O12 <sup>xii</sup> —Na—O12 <sup>xiii</sup>   | 87.13 (15)   |
| O11—Rb1—O11 <sup>iv</sup>                   | 55.43 (15)  | O12 <sup>ii</sup> —Na—O12 <sup>xiii</sup>    | 76.1 (2)     |
| O11 <sup>vii</sup> —Rb1—O11 <sup>iv</sup>   | 84.73 (11)  | O12—Na—O12 <sup>viii</sup>                   | 120.9 (2)    |
| O22 <sup>v</sup> —Rb1—O11 <sup>viii</sup>   | 83.98 (12)  | O12 <sup>xii</sup> —Na—O12 <sup>viii</sup>   | 87.13 (15)   |
| O22 <sup>vi</sup> —Rb1—O11 <sup>viii</sup>  | 152.40 (12) | O12 <sup>ii</sup> —Na—O12 <sup>viii</sup>    | 145.7 (2)    |
| O11—Rb1—O11 <sup>viii</sup>                 | 84.73 (11)  | O12 <sup>xiii</sup> —Na—O12 <sup>viii</sup>  | 87.13 (15)   |
| O11 <sup>vii</sup> —Rb1—O11 <sup>viii</sup> | 55.43 (15)  | O12—Na—O12 <sup>v</sup>                      | 87.13 (15)   |
| O11 <sup>iv</sup> —Rb1—O11 <sup>viii</sup>  | 109.10 (16) | O12 <sup>xii</sup> —Na—O12 <sup>v</sup>      | 145.7 (2)    |
| O22 <sup>v</sup> —Rb1—O12 <sup>v</sup>      | 56.14 (12)  | O12 <sup>ii</sup> —Na—O12 <sup>v</sup>       | 87.13 (15)   |
| O22 <sup>vi</sup> —Rb1—O12 <sup>v</sup>     | 91.18 (12)  | O12 <sup>xiii</sup> —Na—O12 <sup>v</sup>     | 120.9 (2)    |
| O11—Rb1—O12 <sup>v</sup>                    | 95.93 (11)  | O12 <sup>viii</sup> —Na—O12 <sup>v</sup>     | 76.1 (2)     |
| O12—C1—C2—O22                               | 13.4 (10)   | Rb1 <sup>ii</sup> —C1—O11—Rb1 <sup>i</sup>   | 44.6 (12)    |
| O11—C1—C2—O22                               | −168.2 (6)  | O11—C1—O12—Na                                | 24.7 (10)    |
| Rb1—C1—C2—O22                               | 115.8 (6)   | C2—C1—O12—Na                                 | −157.1 (4)   |
| Rb1 <sup>ii</sup> —C1—C2—O22                | −14.0 (6)   | Rb1—C1—O12—Na                                | 78.5 (5)     |
| O12—C1—C2—O21                               | −167.4 (6)  | Rb1 <sup>ii</sup> —C1—O12—Na                 | −124.1 (6)   |
| O11—C1—C2—O21                               | 11.0 (8)    | O11—C1—O12—Rb1 <sup>ii</sup>                 | 148.8 (5)    |
| Rb1—C1—C2—O21                               | −65.0 (7)   | C2—C1—O12—Rb1 <sup>ii</sup>                  | −32.9 (6)    |
| Rb1 <sup>ii</sup> —C1—C2—O21                | 165.3 (5)   | Rb1—C1—O12—Rb1 <sup>ii</sup>                 | −157.41 (18) |
| O12—C1—C2—Rb1 <sup>ii</sup>                 | 27.3 (5)    | O11—C1—O12—Rb1                               | −53.8 (6)    |
| O11—C1—C2—Rb1 <sup>ii</sup>                 | −154.2 (5)  | C2—C1—O12—Rb1                                | 124.5 (6)    |
| Rb1—C1—C2—Rb1 <sup>ii</sup>                 | 129.8 (3)   | Rb1 <sup>ii</sup> —C1—O12—Rb1                | 157.41 (18)  |
| O12—C1—C2—Rb2                               | −33.2 (10)  | O22—C2—O21—Fe                                | 168.6 (6)    |
| O11—C1—C2—Rb2                               | 145.2 (5)   | C1—C2—O21—Fe                                 | −10.6 (7)    |
| Rb1—C1—C2—Rb2                               | 69.2 (7)    | Rb1 <sup>ii</sup> —C2—O21—Fe                 | 124.4 (11)   |
| Rb1 <sup>ii</sup> —C1—C2—Rb2                | −60.5 (5)   | Rb2—C2—O21—Fe                                | −164.6 (2)   |
| O12—C1—O11—Fe                               | 172.3 (5)   | O22—C2—O21—Rb2 <sup>iii</sup>                | 40.5 (10)    |
| C2—C1—O11—Fe                                | −6.1 (6)    | C1—C2—O21—Rb2 <sup>iii</sup>                 | −138.7 (4)   |
| Rb1—C1—O11—Fe                               | 111.3 (3)   | Rb1 <sup>ii</sup> —C2—O21—Rb2 <sup>iii</sup> | −3.7 (16)    |
| Rb1 <sup>ii</sup> —C1—O11—Fe                | −108.2 (9)  | Rb2—C2—O21—Rb2 <sup>iii</sup>                | 67.3 (5)     |
| O12—C1—O11—Rb1                              | 61.0 (6)    | O21—C2—O22—Rb1 <sup>ii</sup>                 | −158.0 (5)   |
| C2—C1—O11—Rb1                               | −117.4 (4)  | C1—C2—O22—Rb1 <sup>ii</sup>                  | 21.1 (9)     |
| Rb1 <sup>ii</sup> —C1—O11—Rb1               | 140.5 (9)   | Rb2—C2—O22—Rb1 <sup>ii</sup>                 | 157.4 (7)    |
| O12—C1—O11—Rb1 <sup>i</sup>                 | −34.9 (9)   | O21—C2—O22—Rb2                               | 44.5 (9)     |

|                             |           |               |            |
|-----------------------------|-----------|---------------|------------|
| C2—C1—O11—Rb1 <sup>i</sup>  | 146.7 (4) | C1—C2—O22—Rb2 | −136.3 (5) |
| Rb1—C1—O11—Rb1 <sup>i</sup> | −95.9 (4) |               |            |

Symmetry codes: (i)  $-z+1/2, -x+1, y-1/2$ ; (ii)  $-z+1, x+1/2, -y+3/2$ ; (iii)  $-y+5/4, -x+5/4, -z+1/4$ ; (iv)  $-y+1, z+1/2, -x+1/2$ ; (v)  $y-1/2, -z+3/2, -x+1$ ; (vi)  $-z+3/4, y+1/4, x-1/4$ ; (vii)  $y-3/4, x+3/4, -z+3/4$ ; (viii)  $z-1/4, -y+7/4, x+1/4$ ; (ix)  $z+1/2, -x+3/2, -y+1$ ; (x)  $-y+3/2, -z+1, x-1/2$ ; (xi)  $x+1/4, z+3/4, -y+3/4$ ; (xii)  $-x+3/4, z+1/4, y-1/4$ ; (xiii)  $-y+5/4, -x+5/4, -z+5/4$ .

### sodium and rubidium tris(oxalate)ferrate(III) (P4132)

#### Crystal data

NaRb<sub>5</sub>[Fe(C<sub>2</sub>O<sub>4</sub>)<sub>3</sub>]<sub>2</sub>

$M_r = 1090.16$

Cubic,  $P4_32$

$a = 13.7995$  (3) Å

$V = 2627.79$  (17) Å<sup>3</sup>

$Z = 4$

$F(000) = 2048$

$D_x = 2.756$  Mg m<sup>−3</sup>

Mo  $K\alpha$  radiation,  $\lambda = 0.71073$  Å

Cell parameters from 1328 reflections

$\theta = 4.4\text{--}27.3^\circ$

$\mu = 10.43$  mm<sup>−1</sup>

$T = 293$  K

Fragment, green

$0.48 \times 0.35 \times 0.25$  mm

#### Data collection

Rigaku Oxford Diffraction Xcalibur, Eos,

Gemini

diffractometer

Radiation source: fine-focus sealed X-ray tube,

Enhance (Mo) X-ray Source

Graphite monochromator

Detector resolution: 16.0604 pixels mm<sup>−1</sup>

$\omega$  scans

Absorption correction: multi-scan

(CrysAlisPro; Rigaku OD, 2015)

$T_{\min} = 0.786$ ,  $T_{\max} = 1.000$

4284 measured reflections

961 independent reflections

814 reflections with  $I > 2\sigma(I)$

$R_{\text{int}} = 0.038$

$\theta_{\max} = 27.0^\circ$ ,  $\theta_{\min} = 3.3^\circ$

$h = -14 \rightarrow 16$

$k = -14 \rightarrow 10$

$l = -17 \rightarrow 16$

#### Refinement

Refinement on  $F^2$

Least-squares matrix: full

$R[F^2 > 2\sigma(F^2)] = 0.032$

$wR(F^2) = 0.068$

$S = 1.02$

961 reflections

68 parameters

0 restraints

Primary atom site location: dual

$w = 1/[\sigma^2(F_o^2) + (0.031P)^2 + 3.9422P]$

where  $P = (F_o^2 + 2F_c^2)/3$

$(\Delta/\sigma)_{\max} < 0.001$

$\Delta\rho_{\max} = 1.02$  e Å<sup>−3</sup>

$\Delta\rho_{\min} = -0.95$  e Å<sup>−3</sup>

Absolute structure: Flack  $x$  determined using

251 quotients  $[(F^+)-(F^-)]/[(F^+)+(F^-)]$  (Parsons *et al.*, 2013).

Absolute structure parameter:  $-0.003$  (10)

#### Special details

**Geometry.** All esds (except the esd in the dihedral angle between two l.s. planes) are estimated using the full covariance matrix. The cell esds are taken into account individually in the estimation of esds in distances, angles and torsion angles; correlations between esds in cell parameters are only used when they are defined by crystal symmetry. An approximate (isotropic) treatment of cell esds is used for estimating esds involving l.s. planes.

#### Fractional atomic coordinates and isotropic or equivalent isotropic displacement parameters (Å<sup>2</sup>)

|     | <i>x</i>    | <i>y</i>   | <i>z</i>   | $U_{\text{iso}}^*/U_{\text{eq}}$ |
|-----|-------------|------------|------------|----------------------------------|
| C1  | 0.1366 (5)  | 0.5569 (4) | 0.1362 (5) | 0.0228 (14)                      |
| C2  | 0.0522 (5)  | 0.5213 (5) | 0.2007 (5) | 0.0256 (15)                      |
| O11 | 0.1218 (3)  | 0.5418 (3) | 0.0448 (3) | 0.0254 (10)                      |
| O12 | −0.0240 (3) | 0.4977 (3) | 0.1533 (3) | 0.0291 (10)                      |

|     |              |             |             |             |
|-----|--------------|-------------|-------------|-------------|
| O21 | 0.2066 (3)   | 0.5967 (3)  | 0.1708 (3)  | 0.0327 (11) |
| O22 | 0.0621 (4)   | 0.5170 (4)  | 0.2882 (4)  | 0.0422 (14) |
| Fe  | −0.01047 (6) | 0.48953 (6) | 0.01047 (6) | 0.0218 (4)  |
| Rb1 | 0.24391 (5)  | 0.50609 (5) | 0.3750      | 0.0291 (2)  |
| Rb2 | −0.07858 (6) | 0.57858 (6) | 0.42142 (6) | 0.0451 (4)  |
| Na  | 0.3750       | 0.6250      | 0.1250      | 0.0234 (12) |

*Atomic displacement parameters (Å<sup>2</sup>)*

|     | $U^{11}$    | $U^{22}$    | $U^{33}$    | $U^{12}$     | $U^{13}$     | $U^{23}$     |
|-----|-------------|-------------|-------------|--------------|--------------|--------------|
| C1  | 0.018 (3)   | 0.024 (3)   | 0.027 (4)   | 0.007 (3)    | −0.001 (3)   | 0.001 (3)    |
| C2  | 0.026 (4)   | 0.034 (4)   | 0.017 (3)   | 0.006 (3)    | 0.003 (3)    | −0.001 (3)   |
| O11 | 0.023 (2)   | 0.032 (3)   | 0.022 (3)   | −0.003 (2)   | 0.002 (2)    | −0.0003 (18) |
| O12 | 0.020 (2)   | 0.042 (3)   | 0.025 (3)   | −0.005 (2)   | 0.0052 (18)  | −0.005 (2)   |
| O21 | 0.022 (3)   | 0.041 (3)   | 0.036 (3)   | −0.003 (2)   | −0.005 (2)   | 0.000 (2)    |
| O22 | 0.032 (3)   | 0.069 (4)   | 0.025 (3)   | −0.001 (3)   | 0.003 (2)    | 0.002 (3)    |
| Fe  | 0.0218 (4)  | 0.0218 (4)  | 0.0218 (4)  | 0.0003 (4)   | −0.0003 (4)  | −0.0003 (4)  |
| Rb1 | 0.0292 (3)  | 0.0292 (3)  | 0.0290 (5)  | 0.0020 (4)   | −0.0047 (3)  | −0.0047 (3)  |
| Rb2 | 0.0451 (4)  | 0.0451 (4)  | 0.0451 (4)  | −0.0124 (4)  | 0.0124 (4)   | −0.0124 (4)  |
| Na  | 0.0234 (12) | 0.0234 (12) | 0.0234 (12) | −0.0014 (15) | −0.0014 (15) | 0.0014 (15)  |

*Geometric parameters (Å, °)*

|                         |           |                                           |             |
|-------------------------|-----------|-------------------------------------------|-------------|
| C1—O21                  | 1.210 (7) | Rb1—O22 <sup>vi</sup>                     | 2.785 (5)   |
| C1—O11                  | 1.295 (8) | Rb1—O11 <sup>vii</sup>                    | 3.006 (4)   |
| C1—C2                   | 1.546 (9) | Rb1—O11 <sup>viii</sup>                   | 3.006 (4)   |
| C2—O22                  | 1.216 (8) | Rb1—O11 <sup>ix</sup>                     | 3.059 (4)   |
| C2—O12                  | 1.280 (7) | Rb1—O11 <sup>x</sup>                      | 3.059 (4)   |
| O11—Fe                  | 2.019 (4) | Rb1—O21 <sup>vi</sup>                     | 3.125 (5)   |
| O11—Rb1 <sup>i</sup>    | 3.006 (4) | Rb1—O21 <sup>viii</sup>                   | 3.357 (5)   |
| O11—Rb1 <sup>ii</sup>   | 3.059 (4) | Rb1—O21 <sup>vii</sup>                    | 3.357 (5)   |
| O12—Fe                  | 1.983 (4) | Rb2—O22 <sup>xi</sup>                     | 2.805 (5)   |
| O12—Rb2 <sup>iii</sup>  | 3.106 (4) | Rb2—O22 <sup>xii</sup>                    | 2.805 (5)   |
| O21—Na                  | 2.439 (4) | Rb2—O12 <sup>xiii</sup>                   | 3.106 (4)   |
| O21—Rb1                 | 3.125 (5) | Rb2—O12 <sup>xiv</sup>                    | 3.106 (4)   |
| O21—Rb1 <sup>i</sup>    | 3.357 (5) | Rb2—O12 <sup>xv</sup>                     | 3.106 (4)   |
| O22—Rb1                 | 2.785 (5) | Na—O21 <sup>vii</sup>                     | 2.439 (4)   |
| O22—Rb2                 | 2.805 (5) | Na—O21 <sup>x</sup>                       | 2.439 (4)   |
| Fe—O12 <sup>iv</sup>    | 1.983 (4) | Na—O21 <sup>xvi</sup>                     | 2.439 (4)   |
| Fe—O12 <sup>v</sup>     | 1.983 (4) | Na—O21 <sup>i</sup>                       | 2.439 (4)   |
| Fe—O11 <sup>v</sup>     | 2.019 (4) | Na—O21 <sup>xvii</sup>                    | 2.439 (4)   |
| Fe—O11 <sup>iv</sup>    | 2.019 (4) |                                           |             |
| O21—C1—O11              | 125.7 (6) | O11 <sup>viii</sup> —Rb1—O11 <sup>x</sup> | 55.34 (16)  |
| O21—C1—C2               | 121.2 (6) | O11 <sup>ix</sup> —Rb1—O11 <sup>x</sup>   | 108.96 (17) |
| O11—C1—C2               | 113.0 (5) | O22 <sup>vi</sup> —Rb1—O21 <sup>vi</sup>  | 56.08 (13)  |
| O21—C1—Rb1 <sup>i</sup> | 82.8 (4)  | O22—Rb1—O21 <sup>vi</sup>                 | 91.06 (13)  |
| O11—C1—Rb1 <sup>i</sup> | 66.1 (3)  | O11 <sup>vii</sup> —Rb1—O21 <sup>vi</sup> | 110.57 (12) |

|                                         |             |                                              |             |
|-----------------------------------------|-------------|----------------------------------------------|-------------|
| C2—C1—Rb1 <sup>i</sup>                  | 123.0 (4)   | O11 <sup>viii</sup> —Rb1—O21 <sup>vi</sup>   | 95.97 (11)  |
| O21—C1—Rb1                              | 53.9 (3)    | O11 <sup>ix</sup> —Rb1—O21 <sup>vi</sup>     | 65.63 (12)  |
| O11—C1—Rb1                              | 154.8 (4)   | O11 <sup>x</sup> —Rb1—O21 <sup>vi</sup>      | 151.23 (11) |
| C2—C1—Rb1                               | 74.1 (3)    | O22 <sup>vi</sup> —Rb1—O21                   | 91.06 (13)  |
| Rb1 <sup>i</sup> —C1—Rb1                | 131.92 (18) | O22—Rb1—O21                                  | 56.09 (13)  |
| O22—C2—O12                              | 125.9 (6)   | O11 <sup>vii</sup> —Rb1—O21                  | 95.97 (11)  |
| O22—C2—C1                               | 120.2 (6)   | O11 <sup>viii</sup> —Rb1—O21                 | 110.57 (12) |
| O12—C2—C1                               | 113.9 (5)   | O11 <sup>ix</sup> —Rb1—O21                   | 151.23 (11) |
| O22—C2—Rb1                              | 41.2 (3)    | O11 <sup>x</sup> —Rb1—O21                    | 65.63 (12)  |
| O12—C2—Rb1                              | 159.2 (4)   | O21 <sup>vi</sup> —Rb1—O21                   | 132.93 (17) |
| C1—C2—Rb1                               | 81.3 (3)    | O22 <sup>vi</sup> —Rb1—O21 <sup>viii</sup>   | 114.16 (14) |
| O22—C2—Rb2                              | 39.9 (3)    | O22—Rb1—O21 <sup>viii</sup>                  | 64.53 (13)  |
| O12—C2—Rb2                              | 94.4 (4)    | O11 <sup>vii</sup> —Rb1—O21 <sup>viii</sup>  | 141.04 (12) |
| C1—C2—Rb2                               | 142.1 (4)   | O11 <sup>viii</sup> —Rb1—O21 <sup>viii</sup> | 40.54 (11)  |
| Rb1—C2—Rb2                              | 79.41 (13)  | O11 <sup>ix</sup> —Rb1—O21 <sup>viii</sup>   | 90.67 (11)  |
| C1—O11—Fe                               | 115.3 (4)   | O11 <sup>x</sup> —Rb1—O21 <sup>viii</sup>    | 90.36 (11)  |
| C1—O11—Rb1 <sup>i</sup>                 | 90.7 (3)    | O21 <sup>vi</sup> —Rb1—O21 <sup>viii</sup>   | 62.26 (16)  |
| Fe—O11—Rb1 <sup>i</sup>                 | 108.83 (17) | O21—Rb1—O21 <sup>viii</sup>                  | 116.94 (14) |
| C1—O11—Rb1 <sup>ii</sup>                | 133.3 (4)   | C1 <sup>vii</sup> —Rb1—O21 <sup>viii</sup>   | 160.09 (13) |
| Fe—O11—Rb1 <sup>ii</sup>                | 106.92 (16) | C1 <sup>viii</sup> —Rb1—O21 <sup>viii</sup>  | 20.95 (13)  |
| Rb1 <sup>i</sup> —O11—Rb1 <sup>ii</sup> | 93.89 (12)  | O22 <sup>vi</sup> —Rb1—O21 <sup>vii</sup>    | 64.53 (13)  |
| C2—O12—Fe                               | 116.4 (4)   | O22—Rb1—O21 <sup>vii</sup>                   | 114.16 (14) |
| C2—O12—Rb2 <sup>iii</sup>               | 133.5 (4)   | O11 <sup>vii</sup> —Rb1—O21 <sup>vii</sup>   | 40.54 (11)  |
| Fe—O12—Rb2 <sup>iii</sup>               | 95.68 (16)  | O11 <sup>viii</sup> —Rb1—O21 <sup>vii</sup>  | 141.04 (12) |
| C1—O21—Na                               | 137.0 (4)   | O11 <sup>ix</sup> —Rb1—O21 <sup>vii</sup>    | 90.36 (11)  |
| C1—O21—Rb1                              | 107.8 (4)   | O11 <sup>x</sup> —Rb1—O21 <sup>vii</sup>     | 90.67 (11)  |
| Na—O21—Rb1                              | 98.10 (14)  | O21 <sup>vi</sup> —Rb1—O21 <sup>vii</sup>    | 116.94 (14) |
| C1—O21—Rb1 <sup>i</sup>                 | 76.2 (4)    | O21—Rb1—O21 <sup>vii</sup>                   | 62.26 (16)  |
| Na—O21—Rb1 <sup>i</sup>                 | 92.25 (14)  | C1 <sup>vii</sup> —Rb1—O21 <sup>vii</sup>    | 20.95 (13)  |
| Rb1—O21—Rb1 <sup>i</sup>                | 158.07 (17) | C1 <sup>viii</sup> —Rb1—O21 <sup>vii</sup>   | 160.09 (13) |
| C2—O22—Rb1                              | 122.0 (4)   | O21 <sup>viii</sup> —Rb1—O21 <sup>vii</sup>  | 178.23 (16) |
| C2—O22—Rb2                              | 124.0 (5)   | O22—Rb2—O22 <sup>xi</sup>                    | 116.26 (7)  |
| Rb1—O22—Rb2                             | 110.97 (18) | O22—Rb2—O22 <sup>xii</sup>                   | 116.26 (7)  |
| O12—Fe—O12 <sup>iv</sup>                | 88.19 (18)  | O22 <sup>xi</sup> —Rb2—O22 <sup>xii</sup>    | 116.26 (7)  |
| O12—Fe—O12 <sup>v</sup>                 | 88.19 (18)  | O22—Rb2—O12 <sup>xiii</sup>                  | 89.16 (13)  |
| O12 <sup>iv</sup> —Fe—O12 <sup>v</sup>  | 88.19 (18)  | O22 <sup>xi</sup> —Rb2—O12 <sup>xiii</sup>   | 131.71 (15) |
| O12—Fe—O11 <sup>v</sup>                 | 161.01 (17) | O22 <sup>xii</sup> —Rb2—O12 <sup>xiii</sup>  | 81.59 (14)  |
| O12 <sup>iv</sup> —Fe—O11 <sup>v</sup>  | 106.36 (18) | O22—Rb2—O12 <sup>xiv</sup>                   | 131.71 (15) |
| O12 <sup>v</sup> —Fe—O11 <sup>v</sup>   | 80.31 (17)  | O22 <sup>xi</sup> —Rb2—O12 <sup>xiv</sup>    | 81.59 (14)  |
| O12—Fe—O11 <sup>iv</sup>                | 106.35 (18) | O22 <sup>xii</sup> —Rb2—O12 <sup>xiv</sup>   | 89.16 (13)  |
| O12 <sup>iv</sup> —Fe—O11 <sup>iv</sup> | 80.31 (17)  | O12 <sup>xiii</sup> —Rb2—O12 <sup>xiv</sup>  | 52.74 (13)  |
| O12 <sup>v</sup> —Fe—O11 <sup>iv</sup>  | 161.01 (17) | O22—Rb2—O12 <sup>xv</sup>                    | 81.59 (14)  |
| O11 <sup>v</sup> —Fe—O11 <sup>iv</sup>  | 88.44 (18)  | O22 <sup>xi</sup> —Rb2—O12 <sup>xv</sup>     | 89.16 (13)  |
| O12—Fe—O11                              | 80.31 (17)  | O22 <sup>xii</sup> —Rb2—O12 <sup>xv</sup>    | 131.71 (15) |
| O12 <sup>iv</sup> —Fe—O11               | 161.01 (17) | O12 <sup>xiii</sup> —Rb2—O12 <sup>xv</sup>   | 52.74 (13)  |
| O12 <sup>v</sup> —Fe—O11                | 106.36 (18) | O12 <sup>xiv</sup> —Rb2—O12 <sup>xv</sup>    | 52.74 (13)  |
| O11 <sup>v</sup> —Fe—O11                | 88.43 (18)  | O21—Na—O21 <sup>vii</sup>                    | 86.98 (16)  |
| O11 <sup>iv</sup> —Fe—O11               | 88.43 (18)  | O21—Na—O21 <sup>x</sup>                      | 76.2 (2)    |

|                                             |             |                                            |              |
|---------------------------------------------|-------------|--------------------------------------------|--------------|
| O22 <sup>vi</sup> —Rb1—O22                  | 95.0 (2)    | O21 <sup>vii</sup> —Na—O21 <sup>x</sup>    | 121.3 (2)    |
| O22 <sup>vi</sup> —Rb1—O11 <sup>vii</sup>   | 83.52 (14)  | O21—Na—O21 <sup>xvi</sup>                  | 145.5 (2)    |
| O22—Rb1—O11 <sup>vii</sup>                  | 152.04 (13) | O21 <sup>vii</sup> —Na—O21 <sup>xvi</sup>  | 76.2 (2)     |
| O22 <sup>vi</sup> —Rb1—O11 <sup>viii</sup>  | 152.04 (13) | O21 <sup>x</sup> —Na—O21 <sup>xvi</sup>    | 86.97 (16)   |
| O22—Rb1—O11 <sup>viii</sup>                 | 83.52 (14)  | O21—Na—O21 <sup>i</sup>                    | 86.97 (16)   |
| O11 <sup>vii</sup> —Rb1—O11 <sup>viii</sup> | 110.46 (16) | O21 <sup>vii</sup> —Na—O21 <sup>i</sup>    | 86.97 (16)   |
| O22 <sup>vi</sup> —Rb1—O11 <sup>ix</sup>    | 84.14 (13)  | O21 <sup>x</sup> —Na—O21 <sup>i</sup>      | 145.5 (2)    |
| O22—Rb1—O11 <sup>ix</sup>                   | 152.50 (13) | O21 <sup>xvi</sup> —Na—O21 <sup>i</sup>    | 121.3 (2)    |
| O11 <sup>vii</sup> —Rb1—O11 <sup>ix</sup>   | 55.34 (16)  | O21—Na—O21 <sup>xvii</sup>                 | 121.3 (2)    |
| O11 <sup>viii</sup> —Rb1—O11 <sup>ix</sup>  | 84.61 (12)  | O21 <sup>vii</sup> —Na—O21 <sup>xvii</sup> | 145.5 (2)    |
| O22 <sup>vi</sup> —Rb1—O11 <sup>x</sup>     | 152.49 (13) | O21 <sup>x</sup> —Na—O21 <sup>xvii</sup>   | 86.97 (16)   |
| O22—Rb1—O11 <sup>x</sup>                    | 84.14 (13)  | O21 <sup>xvi</sup> —Na—O21 <sup>xvii</sup> | 86.97 (16)   |
| O11 <sup>vii</sup> —Rb1—O11 <sup>x</sup>    | 84.61 (12)  | O21 <sup>i</sup> —Na—O21 <sup>xvii</sup>   | 76.2 (2)     |
| O21—C1—C2—O22                               | −13.0 (10)  | Rb1—C1—O11—Rb1 <sup>ii</sup>               | −44.8 (12)   |
| O11—C1—C2—O22                               | 168.7 (6)   | O22—C2—O12—Fe                              | −169.1 (6)   |
| Rb1 <sup>i</sup> —C1—C2—O22                 | −115.7 (6)  | C1—C2—O12—Fe                               | 10.3 (7)     |
| Rb1—C1—C2—O22                               | 14.4 (6)    | Rb1—C2—O12—Fe                              | −124.2 (11)  |
| O21—C1—C2—O12                               | 167.5 (6)   | Rb2—C2—O12—Fe                              | 164.4 (2)    |
| O11—C1—C2—O12                               | −10.8 (8)   | O22—C2—O12—Rb2 <sup>iii</sup>              | −40.6 (10)   |
| Rb1 <sup>i</sup> —C1—C2—O12                 | 64.8 (6)    | C1—C2—O12—Rb2 <sup>iii</sup>               | 138.9 (4)    |
| Rb1—C1—C2—O12                               | −165.1 (5)  | Rb1—C2—O12—Rb2 <sup>iii</sup>              | 4.4 (16)     |
| O21—C1—C2—Rb1                               | −27.4 (5)   | Rb2—C2—O12—Rb2 <sup>iii</sup>              | −67.1 (5)    |
| O11—C1—C2—Rb1                               | 154.3 (5)   | O11—C1—O21—Na                              | −24.5 (10)   |
| Rb1 <sup>i</sup> —C1—C2—Rb1                 | −130.1 (3)  | C2—C1—O21—Na                               | 157.3 (4)    |
| O21—C1—C2—Rb2                               | 32.8 (10)   | Rb1 <sup>i</sup> —C1—O21—Na                | −78.2 (5)    |
| O11—C1—C2—Rb2                               | −145.6 (5)  | Rb1—C1—O21—Na                              | 124.2 (6)    |
| Rb1 <sup>i</sup> —C1—C2—Rb2                 | −69.9 (7)   | O11—C1—O21—Rb1                             | −148.7 (5)   |
| Rb1—C1—C2—Rb2                               | 60.1 (5)    | C2—C1—O21—Rb1                              | 33.2 (6)     |
| O21—C1—O11—Fe                               | −172.2 (5)  | Rb1 <sup>i</sup> —C1—O21—Rb1               | 157.58 (19)  |
| C2—C1—O11—Fe                                | 6.0 (6)     | O11—C1—O21—Rb1 <sup>i</sup>                | 53.7 (6)     |
| Rb1 <sup>i</sup> —C1—O11—Fe                 | −111.3 (3)  | C2—C1—O21—Rb1 <sup>i</sup>                 | −124.4 (6)   |
| Rb1—C1—O11—Fe                               | 107.9 (9)   | Rb1—C1—O21—Rb1 <sup>i</sup>                | −157.58 (19) |
| O21—C1—O11—Rb1 <sup>i</sup>                 | −60.9 (6)   | O12—C2—O22—Rb1                             | 157.6 (5)    |
| C2—C1—O11—Rb1 <sup>i</sup>                  | 117.3 (4)   | C1—C2—O22—Rb1                              | −21.9 (8)    |
| Rb1—C1—O11—Rb1 <sup>i</sup>                 | −140.8 (9)  | Rb2—C2—O22—Rb1                             | −158.5 (8)   |
| O21—C1—O11—Rb1 <sup>ii</sup>                | 35.0 (9)    | O12—C2—O22—Rb2                             | −43.9 (9)    |
| C2—C1—O11—Rb1 <sup>ii</sup>                 | −146.7 (4)  | C1—C2—O22—Rb2                              | 136.6 (5)    |
| Rb1 <sup>i</sup> —C1—O11—Rb1 <sup>ii</sup>  | 96.0 (4)    | Rb1—C2—O22—Rb2                             | 158.5 (8)    |

Symmetry codes: (i)  $-z+1/2, -x+1, y-1/2$ ; (ii)  $-x+1/2, -y+1, z-1/2$ ; (iii)  $y-3/4, -x+1/4, z-1/4$ ; (iv)  $-z, x+1/2, -y+1/2$ ; (v)  $y-1/2, -z+1/2, -x$ ; (vi)  $-y+3/4, -x+3/4, -z+3/4$ ; (vii)  $-y+1, z+1/2, -x+1/2$ ; (viii)  $-z+1/4, y-1/4, x+1/4$ ; (ix)  $-x+1/2, -y+1, z+1/2$ ; (x)  $y-1/4, x+1/4, -z+1/4$ ; (xi)  $-y+1/2, -z+1, x+1/2$ ; (xii)  $z-1/2, -x+1/2, -y+1$ ; (xiii)  $-x-1/4, -z+3/4, -y+3/4$ ; (xiv)  $-y+1/4, x+3/4, z+1/4$ ; (xv)  $z-1/4, y+1/4, -x+1/4$ ; (xvi)  $-x+3/4, -z+3/4, -y+3/4$ ; (xvii)  $z+1/4, -y+5/4, x-1/4$ .